



Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CCE 310 :: Software Development Project-I

Sessional Project Proposal

Project title : PetBondhuBD

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PetBondhuBD

● Abstract

This project proposes the development of **PetBondhuBD**, a cross-platform mobile application built with Flutter that aims to assist pet owners, pet shop owners, and animal lovers in Bangladesh. The app will provide functionalities such as pet profile management, vaccination and health reminders, adoption and lost & found services, emergency contacts, and a community forum. With modules for both pet lovers and shop owners, the system will serve as a one-stop platform for managing pet care, purchasing products, and sharing knowledge. Features like push notifications, GPS-based lost & found, and a scalable architecture will ensure accessibility, reliability, and user-friendliness.

● Objectives

- ❖ Provide a user-friendly platform for pet lovers to create pet profiles, manage health records, and set reminders for vaccinations and medicines.
- ❖ Introduce an adoption and lost & found board to help connect pet lovers and ensure the safety of lost pets.
- ❖ Enable pet shop owners to list and manage their products, making them easily accessible to buyers.
- ❖ Facilitate emergency support with nearby veterinary contact information and hospital locations using Google Maps.
- ❖ Build a community hub for pet lovers to share tips, ask questions, and exchange knowledge.
- ❖ Improve pet care efficiency and owner satisfaction through timely notifications, reminders, and tracking features.

● Problem Statements

In Bangladesh, pet owners often rely on manual methods or scattered resources for pet care. There is no unified platform for managing vaccination schedules, accessing reliable health tips, adopting pets, or finding lost ones. Additionally, pet shop owners lack a dedicated platform to showcase their products directly to pet lovers. This

creates gaps in communication, efficiency, and pet welfare. Hence, there is a need for a **single integrated digital solution** to address these challenges.

● Related Work

- ✓ **PetDesk:** Focuses mainly on veterinary appointment management but lacks local adoption and shop integration.
- ✓ **11Pets:** Provides health records and reminders but does not support community-based lost & found or adoption services.
- ✓ **Pawshake:** Connects pet sitters and owners but does not focus on vaccination reminders or shop modules.
- ✓ **Petfinder:** Focuses heavily on adoption but lacks daily care trackers and personalized health features.

These existing apps address parts of the problem but fail to combine **health management, adoption, e-commerce, and community support** in a single localized solution.

● Scope

The proposed application will be developed using Flutter to support Android and iOS platforms. The system will consist of three major modules:

- **Pet Lover Module** – Users can create pet profiles, manage health/vaccination reminders, track daily care, mark pets as lost, and access tips and community features.
- **Pet Shop Owner Module** – Shop owners can list products (food, toys, medicines), manage orders, and interact with customers.
- **Admin Module** – Provides tools for monitoring adoption posts, managing community content, ensuring data privacy, and generating insights.

The app will integrate Google Maps API for lost & found and clinic location tracking, Firebase Cloud Messaging for push notifications, and a scalable backend (Supabase/MySQL). Future scope includes AI-based health suggestions and a donation feature for stray animals.

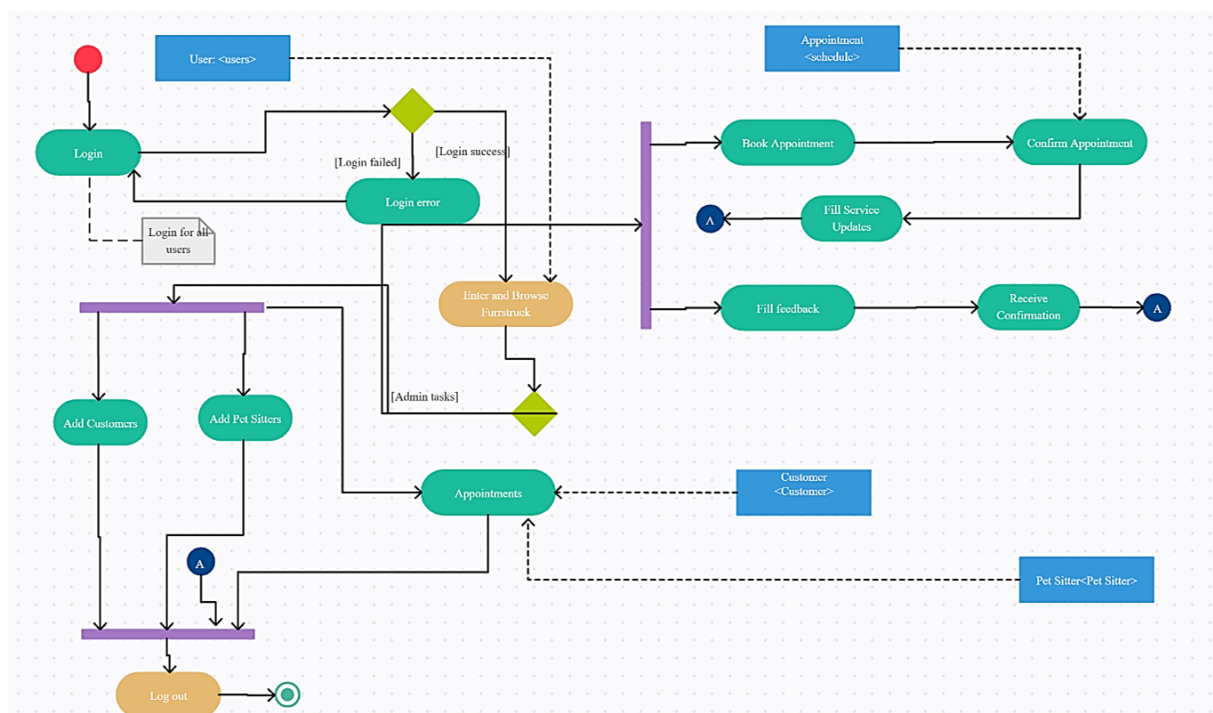
● Working process

The development of PetBondhuBD will follow an **Agile methodology** to ensure flexibility and incremental improvements.

Step	Description
1. Requirement Analysis	Collect user needs (pet lovers, shop owners, admins).
2. UI/UX Design	Design clean and intuitive app interface (Figma).
3. Technology Selection	Flutter (frontend), Firebase/Supabase (backend), MySQL (DB).
4. Database Design	Define ER schema for users, pets, shops, and community data.
5. Frontend Development	Develop Flutter UI for Pet Lover, Shop Owner, and Admin modules.
6. Backend Development	Integrate APIs for authentication, notifications, and GPS features.
7. Testing	Conduct unit and integration testing.
8. Deployment	Publish on Google Play Store and Apple App Store.
9. Feedback & Updates	Collect user feedback and refine features.

● Visual model

◆ Flow Chart Diagram



◆ ER Diagram

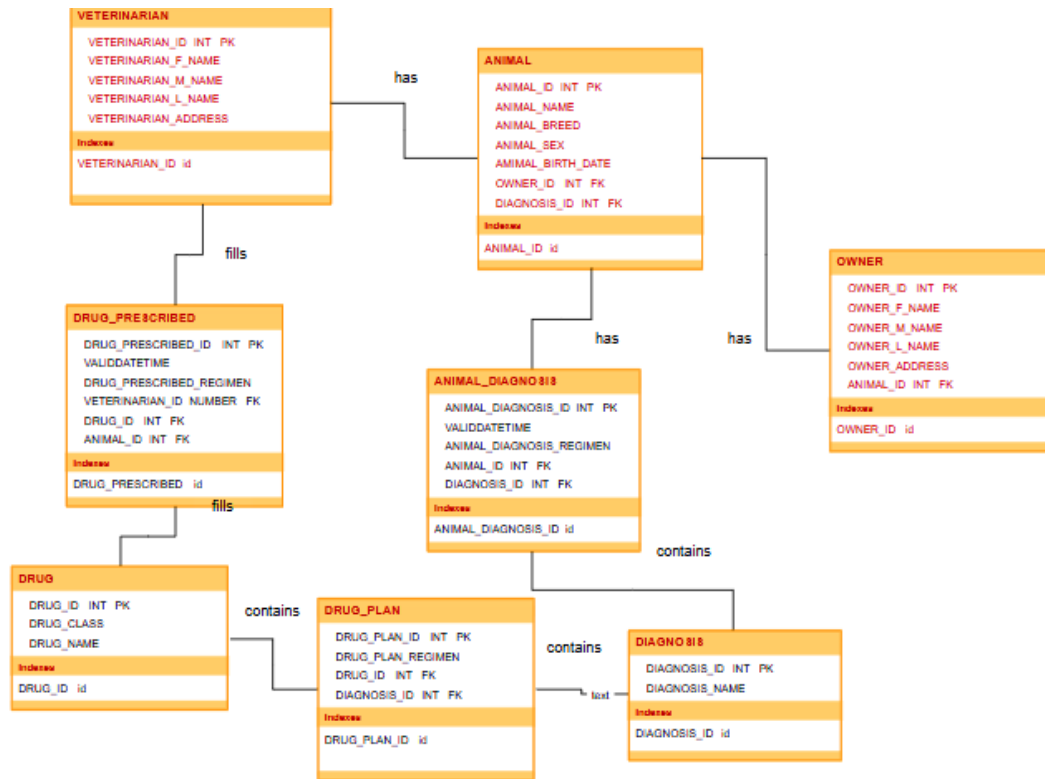


Fig: ER(Entity Relationship) diagram of PetBondhuBD app

◆ Timeline (Gantt Chart)

Task	Week 1-2	Week 3-4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10
Requirements	☑							
UI/UX Design		☑						
Database Design		☑						
Pet Lover Module			☑	☑				
Shop Module				☑	☑			
Admin Panel					☑	☑		
Integration & Testing						☑	☑	
Documentation							☑	☑
Final Submission								☑

Table 2: Development TimeLine PetBondhuBD App

◆ Data Flow Diagram

Level-0 DFD: PetBondhuBD App

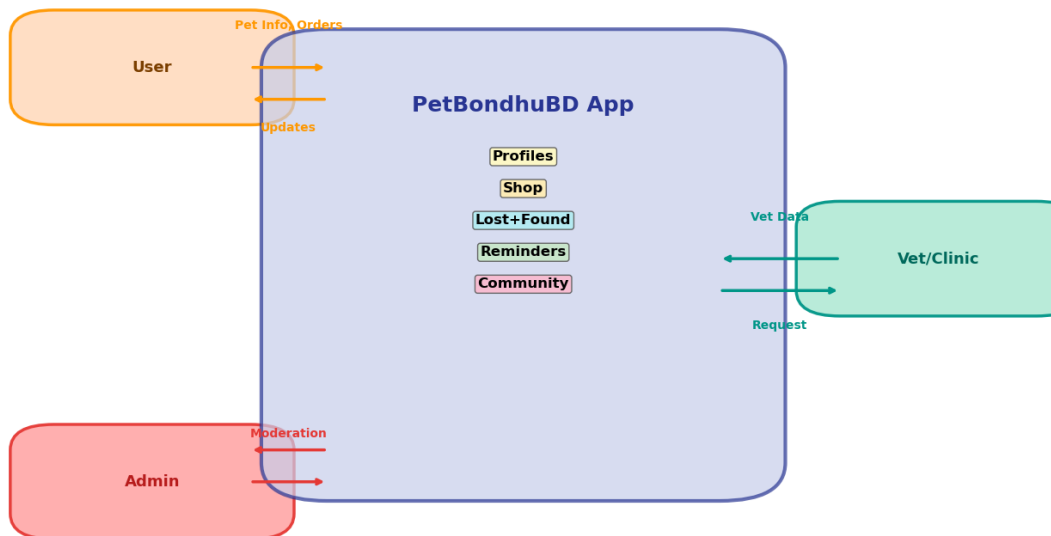
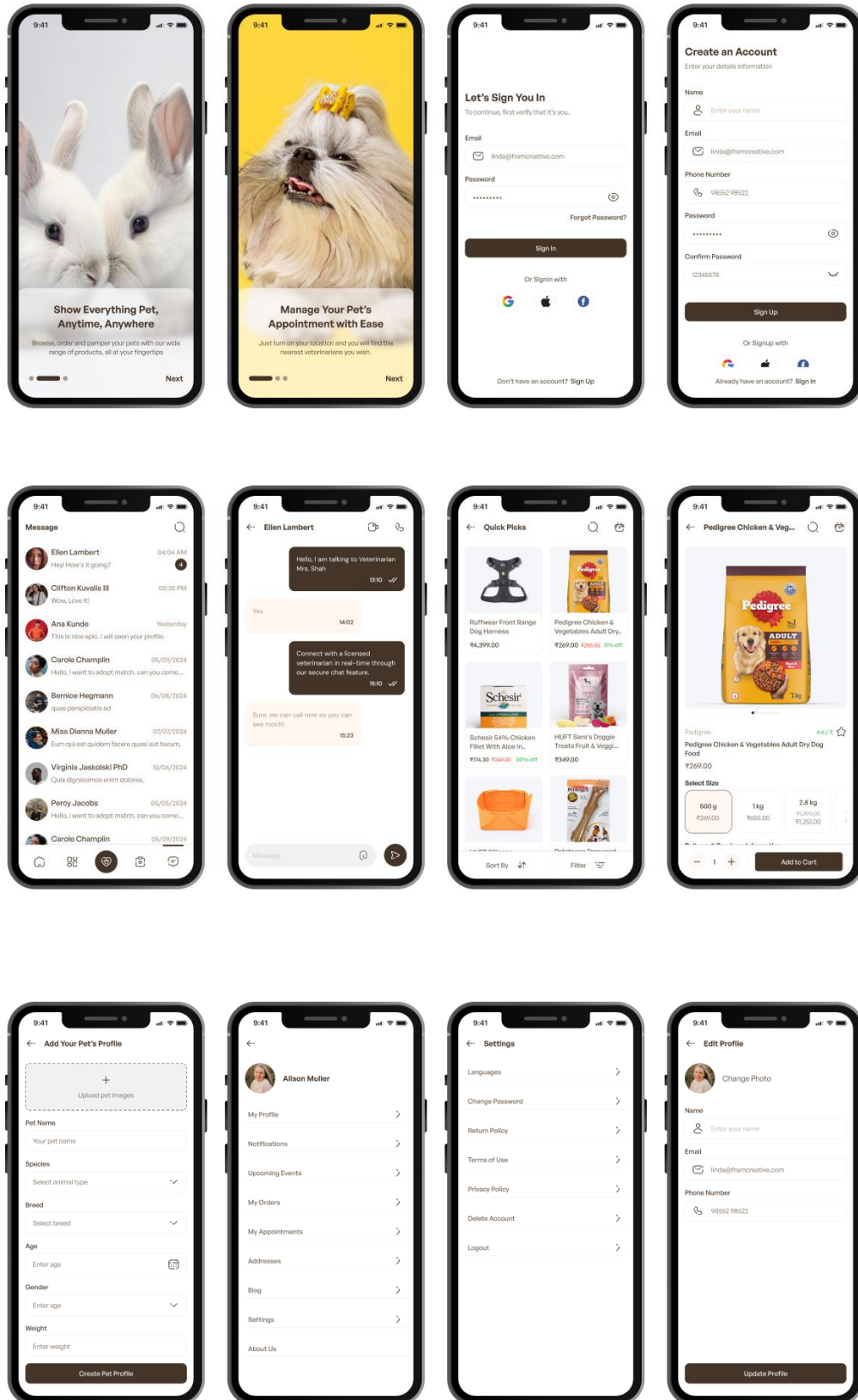


Fig: Data Flow Diagram (DFD) of PetBondhuBD App

◆ UI Mockups



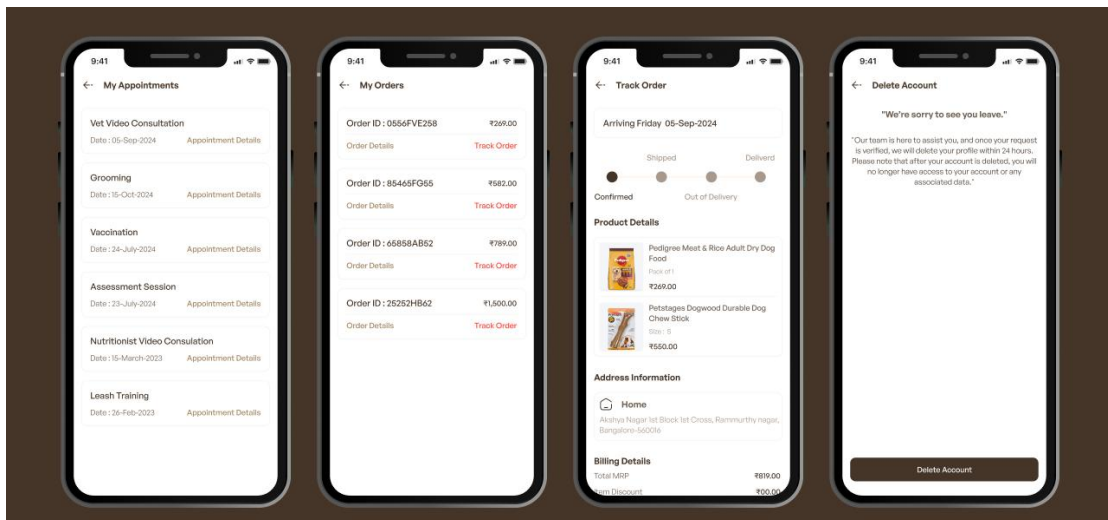
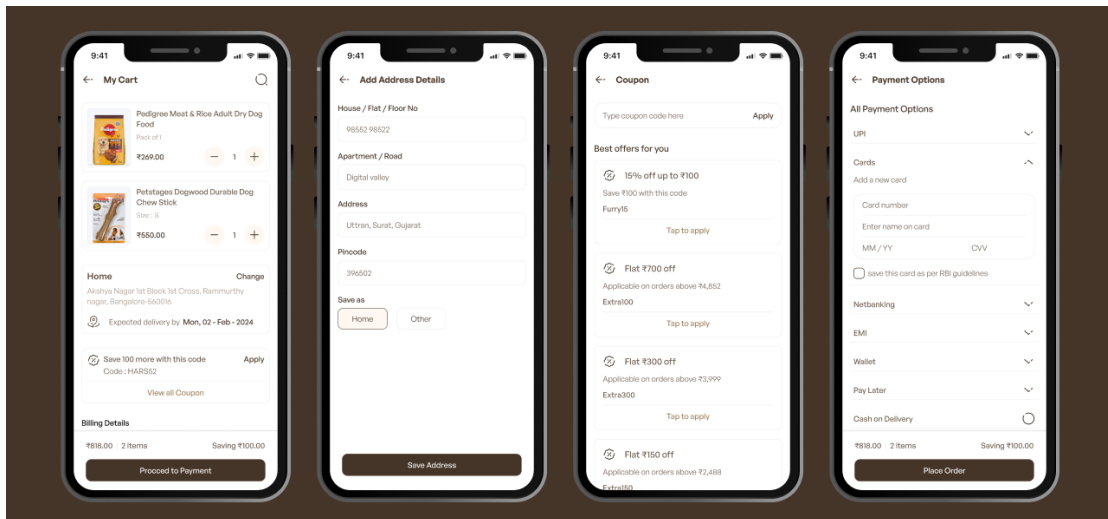
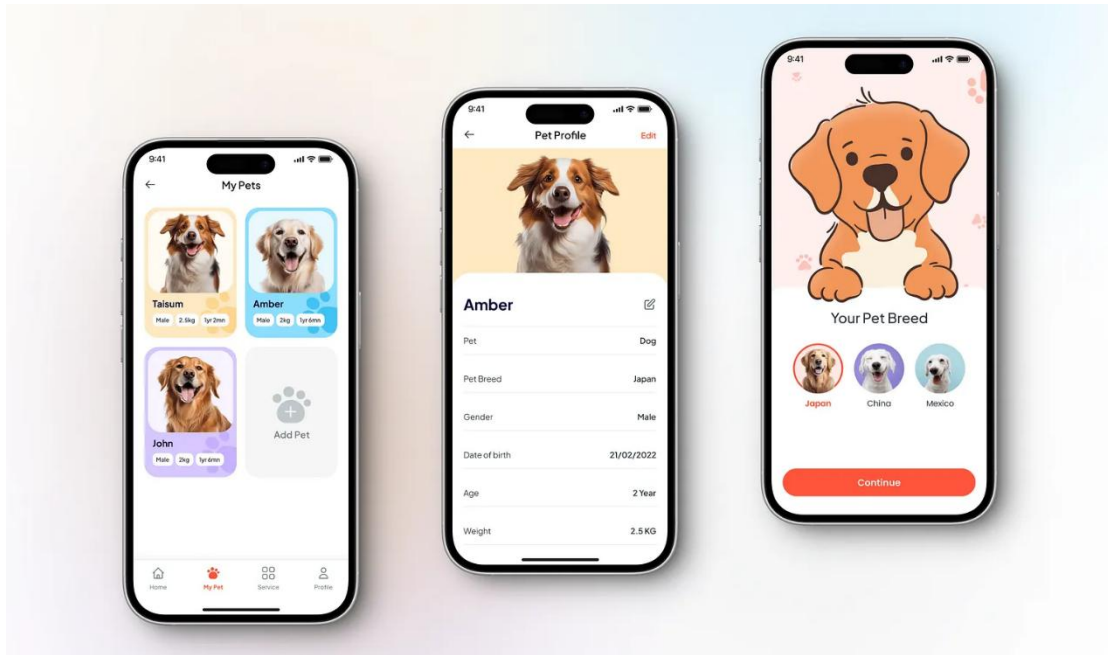


Fig: UI Mock-up of PetBondhuBD app

● Limitations

- ◆ Internet Dependency – App requires stable internet for adoption board, shop, and notifications.
- ◆ Initial Limited Features – Advanced AI health suggestions and donation system will be added later.
- ◆ Geographical Limitation – At first, adoption and shop services may be restricted to selected areas.

● Conclusion

PetBondhuBD is a comprehensive mobile app aimed at improving pet care, adoption, and community building among pet lovers in Bangladesh. With modules for pet lovers and shop owners, it bridges gaps in pet management, health reminders, adoption, and commerce. While initial limitations exist, the project is designed with scalability in mind, ensuring expansion with advanced features in the future.

● References

- Flutter Documentation: <https://docs.flutter.dev/>
- Firebase Documentation: <https://firebase.google.com/docs>
- 11Pets App: <https://www.11pets.com/>
- PetDesk: <https://petdesk.com/>
- Android Studio: <https://developer.android.com/studio>
- Google Maps API: <https://developers.google.com/maps>

Progress table

No.	Date	Objective	status	Remarks
1		Requirement gathering & user needs analysis completed		
2		System architecture, DFD, and ER diagram designed.		
3		UI/UX wireframes and mockups created in Figma.		
4		Database schema (users, pets, shop, orders, etc.) designed.		
5		Flutter frontend core screens implemented.		
6		Backend (Supabase/Firebase) connected with APIs.		
7		Core features (profiles, shop, lost & found, reminders) developed.		
8		GPS (Google Maps) and payment gateway integrated.		
9		Testing, debugging, and user feedback incorporated.		
10		Deployment prepared with full documentation.		